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GE VERNOVA

Gas Power

Gas Fuel Performance Heating
PCHE Heat Exchangers

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The following notices will be found throughout this publication. It is important that the significance of each is thoroughly understood by those using this document. The definitions are as follows:

NOTE

Highlights an essential element of a procedure to ensure correctness.

CAUTION

Indicates a potentially hazardous situation, which, if not avoided, could result in minor or moderate injury or equipment damage.

WARNING

INDICATES A POTENTIALLY HAZARDOUS SITUATION, WHICH, IF NOT AVOIDED, COULD RESULT IN DEATH OR SERIOUS INJURY

*****DANGER*****

INDICATES AN IMMINENTLY HAZARDOUS SITUATION, WHICH, IF NOT AVOIDED WILL RESULT IN DEATH OR SERIOUS INJURY

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I. DESCRIPTION

The thermal performance of the power plant is enhanced by pre-heating the gas fuel before it is burned in the gas turbine with heat taken from the water-steam cycle. There is a dedicated gas fuel heater for each Gas Turbine (GT) and Heat Recovery Steam Generator (HRSG) unit. The gas heater is a counter-flow heat exchanger, and the heating medium is hot feedwater taken from the outlet of the intermediate pressure (IP) and high pressure (HP) economizers of the HRSG. The gas fuel heater discharge water flows to the condensate pump discharge.

The system is configured and operated to improve the performance benefit available from fuel heating. This objective requires that the water leaving the fuel heater and discharging to the condensate system be as cold as possible while the fuel gas to the gas turbine should be as hot as possible. In order to help meet this objective, the fuel heater is controlled to maintain the target fuel temperature.

The heat exchanger is a Printed Circuit Heat Exchanger (PCHE). The core of the PCHE consists of many metal plates that are chemically etched with small channels before being diffusion-bonded together at high pressure and temperature. This results in a core that is a single block of metal with many small gas and water passageways alternating throughout it, in close proximity to each other. Inlet and outlet headers welded to the core allow gas and water to flow through the passageways of the heat exchanger.

The heat exchanger is composed of two stages of fuel heating in a single core:

1. The first stage (IP Section) heats the gas supply utilizing a mixture of IP feedwater and HP feedwater discharged from the second stage.

2. The second stage (HP Section) heats the gas supply from the IP Section of the heater utilizing only HP feedwater.

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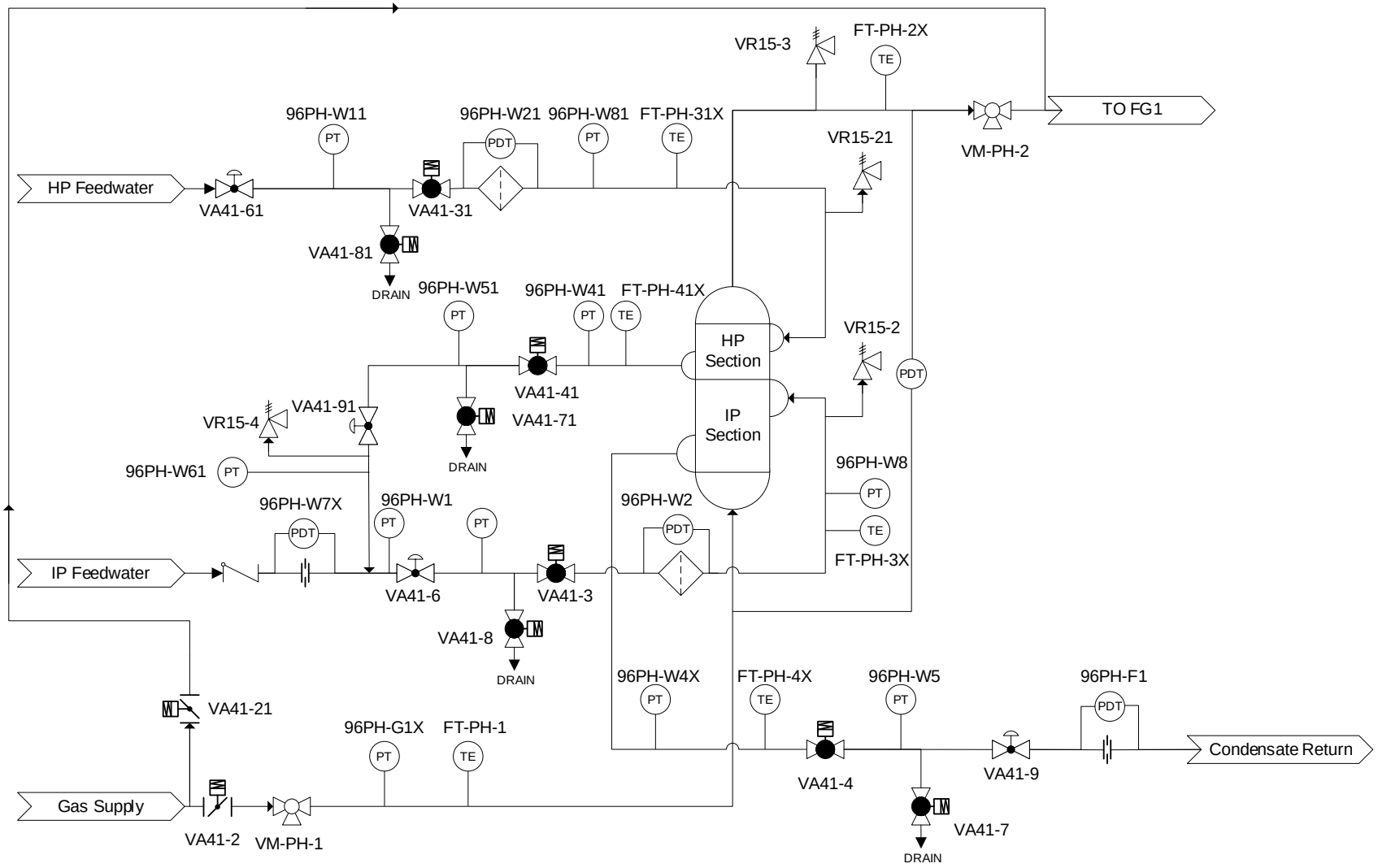


Figure 1. PCHE Performance Heater Arrangement

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II. COMPONENT DESCRIPTION

The below list comprises the main components of the performance heater skid and describes their function. Refer to the gas fuel heating schematic (MLI 4002) for equipment reference.

A. Feedwater Isolation Valves (VA41-3, VA41-4, VA41-31, VA41-41)

In order to isolate the heat exchanger from the feedwater system and condensate systems, the performance heater skid is furnished with isolation valves both upstream and downstream of the heat exchanger. The valve at the inlet of the heat exchanger to isolate the feedwater supply is valve VA41-3 (for the IP Section) and VA41-31 (for the HP Section). The valve at the outlet of the heat exchanger to isolate feedwater from the condensate system is valve VA41-4 (for the IP Section) and VA41-41 (for the HP Section). These valves are metal seated ball valves with pneumatic actuators. When the performance heater is shutdown, both isolation valves are closed to isolate the heat exchanger from the upstream and downstream water systems.

B. Section Feedwater Throttling/Startup Valve (VA41-6, VA41-61)

Prior to initiating water flow to the heat exchanger, the pneumatically actuated globe valves (VA41-6 for the IP Section and VA41-61 for the HP Section) are partially opened. VA41-6 and VA41-3 are used to provide double block and bleed isolation of the IP Section Feedwater Supply; VA41-61 and VA41-31 are used for the same function on the HP Section.

This provides protection for the feedwater supply system from gas migration, in the case of a heat exchanger leak while the heat exchanger is offline. The throttling valves are controlled by digital valve controllers (90VA41-6 for VA41-6 and 90VA41-61 for VA41-61) which receives position command signals from the gas turbine control system and sends valve position feedback to the control system.

C. IP Section Temperature Control Valve (VA41-9)

The IP Section Temperature Control Valve (VA41-9) is a pneumatically actuated globe valve configured to control the flow of feedwater through the IP Section of the heat exchanger. During operation, water from the outlet of this valve will exit the performance heater skid and return to the condensate system for recirculation back to the HRSG. The Feedwater Temperature Control Valve (VA41-9) is used in combination with the Feedwater Outlet Isolation Valve (VA41-4) and the bleed valve (VA41-7) to provide double block and bleed isolation of the return line to the condensate system. This eliminates the potential of gas migrating to the condensate system when the heat exchanger is offline. The temperature control valve is controlled by a digital valve controller (90VA41-9) which receives a position command signal from the gas turbine control system and sends the actual valve position feedback to the control system.

D. HP Section Feedwater Temperature Control Valve (VA41-91)

The HP Section Temperature Control Valve (VA41-91) is a pneumatically actuated globe valve configured to control the flow of feedwater through the HP Section of the heat exchanger. During operation, water from the outlet of this valve will be mixed with the incoming IP Feedwater supply and sent to the IP Section of the heat exchanger. The Feedwater Temperature Control Valve (VA41-91) is used in combination with the Feedwater

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Outlet Isolation Valve (VA41-41) and the bleed valve (VA41-17) to provide double block and bleed isolation HP discharge line. This eliminates the potential of gas migrating to the water system when the heat exchanger is offline. The temperature control valve is controlled by a digital valve controller (VA41-91) which receives a position command signal from the gas turbine control system and sends the actual valve position feedback to the control system.

E. Feedwater Pressure Relief Valve

Pressure relief valves (VR15-2 for the IP Section and VR15-21 for the HP Section) are installed on the water side of the system to protect the heat exchanger from over-pressurization when the feedwater isolation valves (VA41-3 and VA41-4 for the IP Section or VA41-31 and VA41-41 for the HP Section) are closed. The relief valves are set at the design pressure of the respective feedwater system design pressure. The relief valve is sized to protect the heat exchanger and piping on the skid from thermal effects (i.e. heating of trapped water). The relief valve is not sized to protect against over-pressurization of the feedwater supply system. Feedwater over-pressurization protection shall be provided upstream of the Fuel Gas Performance Heater skid by the Plant Integrator (EPC, partner, or customer).

The HP Outlet to IP Inlet relief valve (VR15-4) is sized to protect the IP Section of the system from over-pressurization due to the higher operating pressures in the HP Section. The relief valve is set at the design pressure of the IP Section and sized to account for the scenario where the HP Section Feedwater Temperature Control Valve is fully open with the upstream and downstream sides at their respective design conditions.

For all applications, the vent/drain connections from the skid (such as the discharge from VA41-8) and the relief valve discharge piping shall not be tied together. No isolation valves or other restrictions are allowed downstream of the relief valves.

F. IP Feedwater Inlet Temporary Strainer

A conical “witch hat” strainer is installed at the IP feedwater supply connection upstream of the heat exchanger to prevent debris from damaging the heat exchanger during initial commissioning of the system. A local differential pressure indicator is provided to measure the pressure drop across the strainer to indicate when it is clogged with debris. The operator shall verify that the strainer is installed and monitor the pressure drop across the strainer when the heat exchanger is being commissioned. Refer to the Maintenance section for more information. The strainer is to be removed once the feedwater system is deemed to be clean.

A temporary strainer is not installed on the HP section of the heat exchanger as the skid is configured to reduced flanged connections due to the high temperature and high pressure application.

G. Feedwater Inlet Permanent Filter

A permanent cartridge filter is installed on the IP Section and HP Section of the heat exchanger. The filter is located upstream of the respective heat exchanger section and downstream of the inlet double block and bleed cavity to provide filtration based on the requirements defined by the PCHE manufacturer. Due to the requirements of the PCHE, the feedwater filter provides continuous filtration to protect the PCHE after commissioning. A differential pressure transmitter is provided to continuously monitor the pressure drop across

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the filter and alarm when high pressure drop is observed. Refer to the Maintenance section for more information.

H. Feedwater Flow Meters

Two orifice flow meters are included on the performance heater skid. The IP inlet flow meter is sized to measure the IP flow rate to the IP section of the heat exchanger prior to mixing with the discharged HP Feedwater. The total feedwater flow meter is used to measure the total water flow rate (combined IP and HP) through the IP heat exchanger. Each flowmeter is equipped with differential pressure transmitters (96PH-W7X for the IP inlet flow meter and 96PH-F1 for the combined IP and HP flow meter) to measure the pressure drop across the orifice and determine each flow meters respective flow rate.

I. Automated Fuel Gas Isolation and Bypass Valves

An automated isolation valve (VA41-2) and bypass valve (VA41-21) is provided to allow gas fuel to flow through the heat exchanger during performance heater operation or to flow gas around the heat exchanger when in “Shutdown State”. The gas fuel automated isolation and bypass valves are pneumatically actuated butterfly valves equipped with a positioner (90VA41-2 and 90VA41-21) and trip solenoids (20VA41-2 and 20VA41-21) to command the valves to the desired position based on the performance heater operating mode.

J. Manual Fuel Gas Isolation Valves

In order to isolate the heat exchanger from the gas fuel system, manual isolation valves (VM-PH-1 and VM-PH-2) are installed at the inlet and outlet of the gas side of the heat exchanger. These manual valves are ball valves with integrated double block and bleed sections with incorporated vents.

K. Gas Fuel Pressure Relief Valve

Pressure relief valve (VR15-3) is installed on the gas side of the system to protect the heat exchanger from over-pressurization when the gas isolation valves are closed. The relief valve is set at the gas system design pressure. The heat exchanger vent piping and the relief valve discharge piping shall not be tied together, and no isolation valve is allowed downstream the relief valve. The relief valve is sized to protect the heat exchanger and piping on the skid from thermal effects (i.e. heating of trapped gas). The relief valve is not sized to protect against over-pressurization of the gas fuel supply system. Fuel gas supply over-pressurization protection shall be provided upstream of the Fuel Gas Performance Heater skid by the Plant Integrator (EPC, partner, or customer).

III. OPERATING MODES

A. Shutdown State

Prior to enabling the heat exchanger, the water side of the heater is not pressurized and the double block and bleed is active on the water inlet and outlet side. The normal position of all valves in the system during a shutdown are defined in Table 1. For subsequent operating modes, the valves will start from the referenced state.

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Table 1. Shutdown State Valve Positions

Valve	Position	Description/Comments
VM-PH-1	Open	Gas Inlet Manual Valve - Locked Open
VM-PH-2	Open	Gas Outlet Manual Valve - Locked Open
VA41-6	Closed	IP Water Inlet Control Valve
VA41-3	Closed	IP Water Inlet Isolation Valve
VA41-8	Open	IP Water Inlet Bleed Valve
VA41-4	Closed	IP Water Outlet Isolation Valve
VA41-9	Closed	IP Water Outlet Control Valve
VA41-7	Open	IP Water Outlet Bleed Valve
VA41-61	Closed	HP Water Inlet Control Valve
VA41-31	Closed	HP Water Inlet Isolation Valve
VA41-81	Open	HP Water Inlet Bleed Valve
VA41-41	Closed	HP Water Outlet Isolation Valve
VA41-91	Closed	HP Water Outlet Control Valve
VA41-71	Open	HP Water Outlet Bleed Valve
VA41-2	Closed	Gas Fuel Isolation Valve
VA41-21	Open	Gas Fuel Bypass Valve

B. GT/HRSG Start-up Mode

During the pre-start checks, the gas fuel isolation valve (VA41-2) and bypass valve (VA41-21) will be commanded to their energized state. An alarm will be initiated if the valves do not stroke within the specified time limits. After the pre-start check is completed, VA41-2 will open and VA41-21 will close to enable operation of the gas fuel performance heater during gas turbine operation.

The performance heater startup pre-fill sequence begins when a GT start on gas fuel is selected. The performance heater begins the IP Section pressurization process by commanding the IP feedwater inlet control valve (VA41-6) to its minimum stroke position. This requires the feedwater pumps to be running as part of the plant permissive to start. The inlet control valve minimum stroke position is used during the startup sequence to address water hammer concerns and purge any remaining gas through the vent lines.

Once the IP feedwater inlet control valve (VA41-6) is at minimum stroke, the IP feedwater inlet bleed valve (VA41-8) is closed. No further action is taken until the IP feedwater inlet double block and bleed cavity pressure transmitters (96PH-W1X) confirm that water pressure is higher than the gas inlet pressure (96PH-G1X). If this is not satisfied, then startup of the performance heater is aborted, and the valves are returned to their Shutdown State.

Once sufficient feedwater pressure is established, the IP feedwater inlet isolation valve (VA41-3) is commanded open, allowing water to fill the IP Section water side of the heat exchanger. After confirming VA41-3 is open, the IP feedwater outlet isolation valve (VA41-4) is commanded open, which purges any trapped gas through the outlet bleed valve (VA41-7).

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Once VA41-4 has been confirmed open, the bleed valve (VA41-7) will close. After VA41-7 is confirmed closed, the IP feedwater inlet control valve (VA41-6) will move to the fully open position.

Once the outlet cavity (between VA41-4 and VA41-9) is pressurized, the comparison of water pressure and gas pressure uses the feedwater outlet pressure transmitters (96PH-W4X) for all further operating modes. If this requirement is not met at any point during operation, the performance heater will isolate feedwater and return to the Shutdown State. If the pressure requirement is met, the system will remain in this position until the temperature control valve (VA41-9) is commanded to the minimum stroke position.

Once flow is established through the IP Section of the heat exchanger, the HP Section of the heat exchanger skid will follow the same pressurization sequence using the HP valves with the intent to reduce water hammer effects. This result in opening the HP Section isolation valves and closing the vent valves. Water flow is then initiated by opening the HP Temperature control valve (VA41-91) to a minimum flow control position.

If the gas turbine does not ignite, the water valves may be kept open (except VA41-9 and VA41-91) in anticipation for another start attempt. If another start attempt is not planned the operator can select the stop button, which will return the water side of the heat exchanger to double block and bleed isolation (returning the system to Shutdown State). If the gas turbine trips after flame is established, the heater will revert to Shutdown State.

Two temperature control loops are utilized for control of the performance heater. One temperature control loop modulates the IP temperature control valve (VA41-9) and the other modulates the HP temperature control valve (VA41-91).

The IP temperature control valve (VA41-9) is modulated by the turbine control system to maintain a constant cold end pinch offset (KTCPH3_OFFSET). In addition to meeting this requirement, there is a minimum IP flow rate, prior to mixing with feedwater discharged from the HP Section of the heat exchanger, that must be maintained during operation (KMBP36CF001L). The IP temperature control valve will modulate to the required position to meet both requirements as shown in Figure 2.

The HP temperature control valve (VA41-91) will begin modulating to target the setpoint defined by “DCS Startup Setpoint” and increase at the rate set in KGFPHTEMP_SP_RINC. As the system continues warming up, the hot end pinch offset (KTCPH2_OFFSET) will be used to limit feedwater flow.

The HP temperature control valve (VA41-91) is modulated to maintain a constant hot end pinch offset (KTCPH2_OFFST) based on the differential between the feedwater inlet temperature (HP feedwater side of heater inlet) and the fuel gas outlet temperature (gas side heater outlet) while fuel flows and feedwater temperatures increase. This is balanced with ensuring the minimum IP feedwater flowrate (KMBP36CF001L) is maintained while anticipating increased demand from the FSR feedforward function. The HP temperature control valve will modulate to the required position to meet all requirements as shown in Figure 3.

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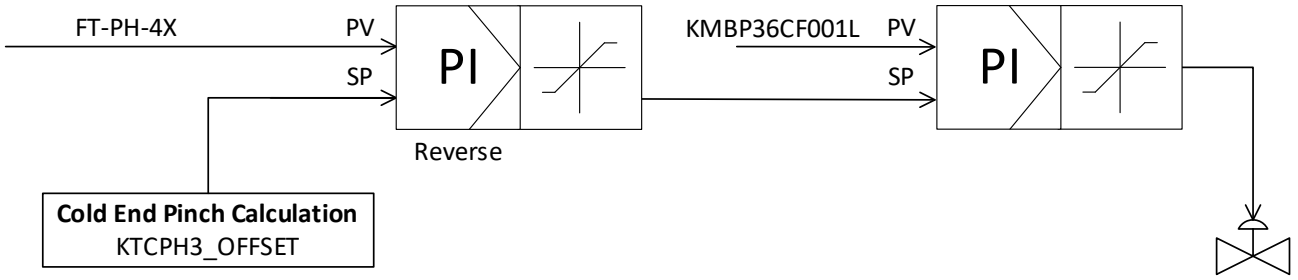


Figure 2. IP Temperature Control Valve (VA41-9) Temperature Control Loop

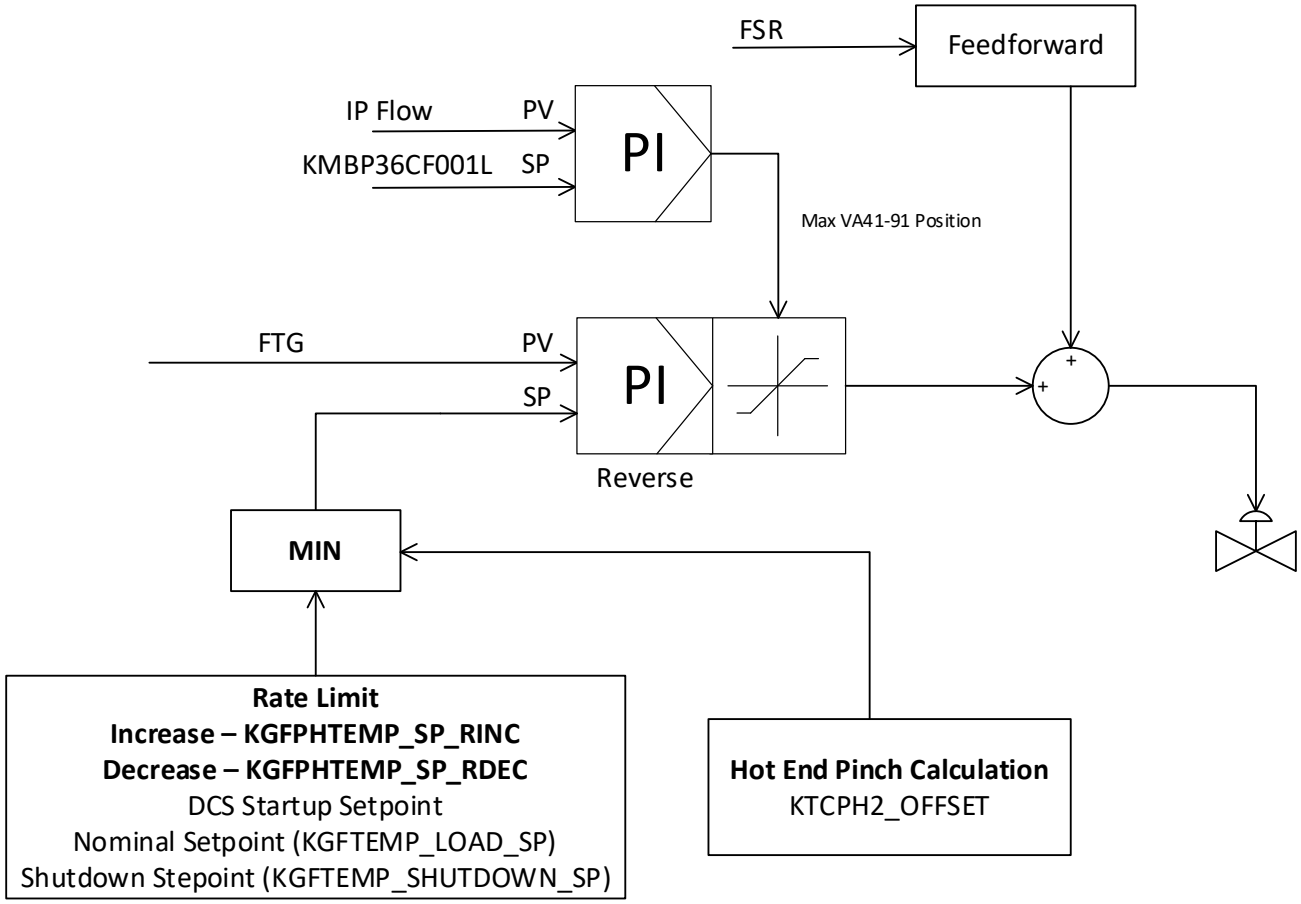


Figure 3. HP Temperature Control Valve (VA41-91) Temperature Control Loop

C. Normal Operation Mode

As the GT/HRSG load is changed during normal plant operation in the higher load range, the gas fuel temperature control will modulate the HP Temperature Control Valve (VA41-91) to control the gas fuel temperature to the setpoint (KGFTEMP_LOAD_SP) while modulating the IP Temperature Control Valve (VA41-9) to maintain minimum IP feedwater flow. In the lower load range, when the HP economizer outlet temperature is insufficient, feedwater temperature is incapable of maintaining the setpoint gas temperature; the gas temperature is gradually decreased using the hot end pinch offset (KTCPH2_OFFSET).

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D. GT/HRSG Shutdown

During normal gas turbine shutdown, gas fuel temperature continues to be controlled by either the setpoint temperature or hot end pinch control until the gas turbine is shutdown. It is required that sufficient feedwater pressure be available to maintain heater operation until the gas turbine is shut down.

When the unit is shut down, the feedwater section of the heat exchanger will revert to the Shutdown State. This is intended to vent gas fuel through the double block and bleed cavities in the event of a leak in the heat exchanger.

E. Shutdown State

When a protection is triggered during operation of the gas fuel performance heater, the system will revert to the Shutdown State. During these events, the isolation sequence is as follows:

1. Close all outlet isolation valves (VA41-4, VA41-9, VA41-41, and VA41-91).
2. Opening outlet bleed valves (VA41-7 and VA41-71) after confirmation that the outlet isolation valves are closed.
3. Close inlet isolation valves (VA41-3, VA41-6, VA41-31, and VA41-61) at the same time outlet bleed valve is commanded open.
4. Open inlet bleed valves (VA41-8 and VA41-81) after confirmation that the inlet isolation valves are closed.

When the protection is triggered, the bypass valve (VA41-21) will open. Once the bypass valve is open, the inlet isolation valve (VA41-2) will close flowing gas around the heat exchanger.

In the event of the performance heater entering the Shutdown State, the event that resulted in heat exchanger isolation shall be identified. During the next shutdown, the cause of the event shall be identified and addressed. The gas fuel performance heater cannot be restarted until the gas turbine has shutdown.

CAUTION

The gas turbine must shut down to enable fuel heating after entering Shutdown State. The event that resulted in heat exchanger isolation shall be determined and addressed to prevent the same issue from occurring during the next operating period.

The gas turbine can continue running on cold fuel after these events within limits determined by the combustion system. Refer to the separate sections on Equipment Protection for more information on the scenarios that result in a Shutdown State being activated during operation.

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F. HP Section Isolation

There are protections that can be initiated during operation of the gas fuel performance heater which will result in only the HP Section of the heat exchanger skid being isolated from feedwater. When this condition is triggered, the HP section of the heat exchanger will be isolated, but IP feedwater will continue flowing through the IP PCHE section and gas fuel will flow through the IP and HP section of the heat exchanger to retain a level of fuel heating. Below is the isolation sequence:

1. Close all outlet isolation valves (VA41-41, and VA41-91).
2. Opening outlet bleed valve (VA41-71) after confirmation that the outlet isolation valves are closed.
3. Close inlet isolation valves (VA41-31, and VA41-61) at the same time outlet bleed valve is commanded open.
4. Open inlet bleed valve (VA41-81) after confirmation that the inlet isolation valves are closed.

The cause of the fault that resulted in HP Section isolation should be determined. Once the cause of the fault has been determined and corrected, the system can be reset, and HP fuel heating will resume.

G. Gas Fuel to Liquid Fuel Transfer

In the event of a Gas-to-Liquid fuel transfer, the fuel gas performance heater will isolate feedwater through the same sequence followed during a GT/HRSG Shutdown.

H. Liquid Fuel to Gas Fuel Transfer

Once the GT has successfully transferred to 100% gas fuel operation, the performance heater will be enabled. The startup sequence of the performance heater follows the sequence as described for GT/HRSG Startup Mode and Normal Operation Mode.

NOTE

Startup sequences should not be modified as this may result in a system lockout or damage to equipment. Refer to GEK111309 for additional recommendations.

I. Load Rejection and Step to Spinning Reserve

For transient protections, such as load rejection or step to spinning reserve, there is an override of the temperature control valve setpoint intended to reduce the fuel temperature overshoot. Once a load rejection is detected, the output of the temperature control valve is set to a lower position for a set time, then released back to active temperature control.

J. Gas Turbine Trip

After a gas turbine trip, the feedwater isolation valves are commanded closed in the same sequence described in the Shutdown State section. The gas fuel flow through the

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performance heater will be stopped when the downstream gas stop and control valves are closed (see MLI 0422).

K. Gas Fuel Performance Heater OFF

Selecting the gas fuel performance heater “OFF” button on the HMI screen will allow the gas turbine to run on cold fuel and keep the feedwater isolation valves closed, the gas fuel inlet isolation valve (VA41-2) closed, and the gas fuel bypass valve (VA41-21) open during gas turbine operation. The operator can choose to start the gas turbine and run on cold fuel if there is a known problem with the PCHE or if there is likely a problem that will result in the heater entering a Shutdown State during operation. This feature can also be used during plant commissioning to start the gas turbine on cold fuel so the plant can commission the feedwater system prior to introducing feedwater flow to the PCHE. If the gas turbine is started with the heater “OFF”, then the unit must be shutdown and on turning gear to enable the performance heater.

IV. EQUIPMENT PROTECTION

A. Heat Exchanger Overpressure Protection

The fuel gas performance heater is equipped with pressure safety valves for the gas and water sides:

1. A pressure relief valve is provided on the gas side to avoid overpressure due to gas thermal expansion at no flow.

2. A pressure relief valve is provided on the water sides to avoid overpressure due to water thermal expansion or from vapor generation due to a pool fire on the skid at no flow.

B. Low Temperature (Freeze) Protection

The PCHE, water piping, and drains shall be heat traced and insulated on all projects where the site ambient temperature may drop below freezing. The water temperature will be monitored by the turbine control panel at all times when the unit is shutdown, and an alarm will be annunciated if the PCHE water temperature measured by FT-PH-4X is below 38 °F. The operator action in response to this alarm is to check the functionality of the PCHE heat tracing and insulation. Water freezing in the PCHE cavity when the isolation valves are closed may result in damage to the PCHE, piping, and/or valves. The alarm is intended to bring attention to the operator before reaching the freezing point.

C. High Gas Temperature Protection

Typically, the gas fuel system design temperature is 25 °F above the gas fuel setpoint temperature (i.e. for 600 °F setpoint, the design temperature will be 625 °F). In order to limit the time the gas fuel temperature exceeds the design temperature, there are two protections that are established.

1. When the temperature control loop is in regulation, the HP feedwater control valve VA41-91 will be ramped closed if the PCHE gas outlet temperature (FT-PH-2X) is 25 °F above the setpoint (i.e. 625 °F for a 600 °F unit). The intent is to prevent the gas temperature measured at the gas fuel module from reaching the design temperature and resulting in the performance heater being placed in a HP Section Isolation (see below).

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2. The gas fuel performance heater will go into HP Section Isolation during any operating case if the gas fuel temperature at the inlet of the gas fuel module measured by FTG-1x (see MLI 0422) is 25 °F above the setpoint (i.e. 625 °F for a 600 °F unit).

The two levels of protection for over temperature are provided as a means for system level protection without the need to trip the gas turbine.

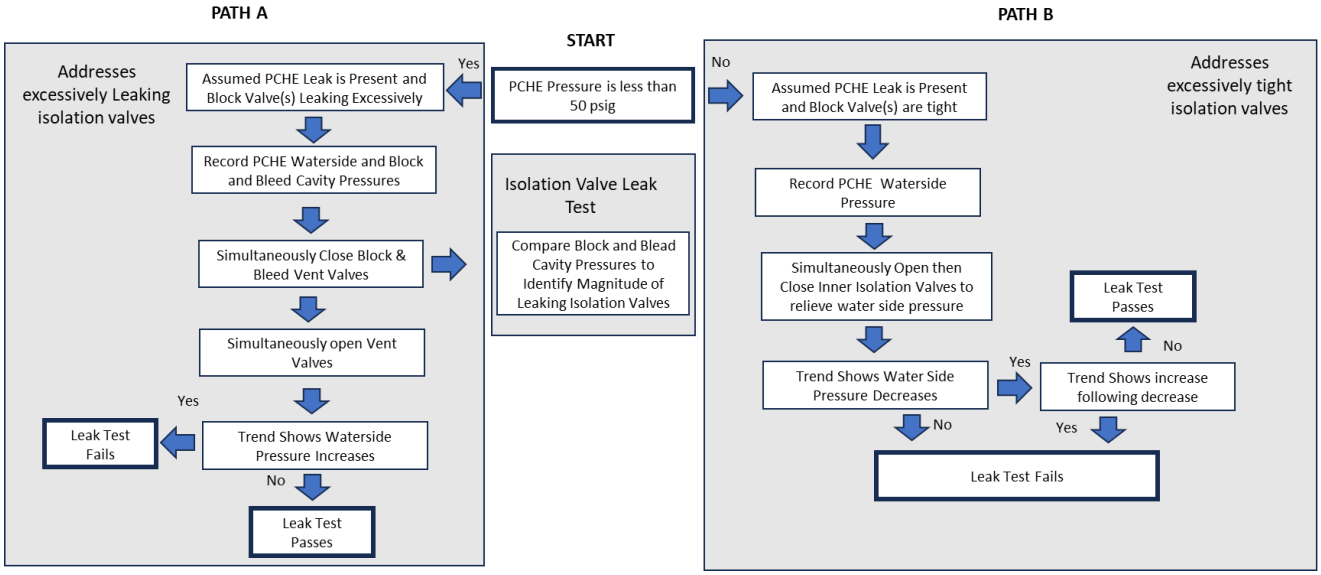
D. Performance Heater Leakage Detection and Protection

A manually initiated leak test is implemented as part of the performance heater software that will be available to be performed following a GT shutdown and after the feedwater system has been turned off. This is designed to determine if a leak has developed during previous operation of the performance heater. For the PCHEs that utilize both Intermediate Pressure (IP) and High Pressure (HP) feedwater for fuel heating, the two sections are tested independently but sequentially. The following are the details of the leak testing process:

- GT has been off-line a minimum of 6 hours.
- PCHE is in SAFE MODE. (Inlet and Outlet water side double block and bleed is enabled)
- Gas pressure in excess of 350 psig is present within the PCHE
- Feedwater water system is in shutdown mode (No IP feedwater pressure to the fuel heating skid).
- Inlet and outlet block and bleed cavities for both IP and HP are less than 5 psig. (This ensures proper double block and bleed isolation)
- Manually initiate leak test through HMI.

IP Section

- IP PCHE water side pressure as measured is measured. If IP pressure is less than 50 psig, **Path A** is executed. If pressure is greater than 50 psig, **Path B** is executed.



- **Path A**

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- IP Inlet and Outlet Double Block and Bleed cavity pressures are recorded.
- IP Inlet and Outlet Double Block and Bleed Vent Valves are simultaneously closed.
- Valves kept closed for 5 seconds then command open.
- IP Inlet and Outlet Double Block and Bleed cavity pressures from the point of each vent valves being sensed closed to the point at which they begin to open are measured. If a pressure rise of 5 psi or greater is detected in either the IP Inlet Double Block and Bleed cavity or IP Outlet Double Block and Bleed cavity during this time the IP leak test FAILS.
  - A leak identified by Path A indicates one or both of the IP the water side isolation valves is leaking excessively. An ANUNCIATION is activated identifying which or if both valves are leaking excessively. Maintenance should be performed on these valves.
- If a pressure rise of less than 5 psi is seen in either the IP Inlet Double Block and Bleed or IP Outlet Double Block and Bleed cavities is detected, the IP leak test PASSES.

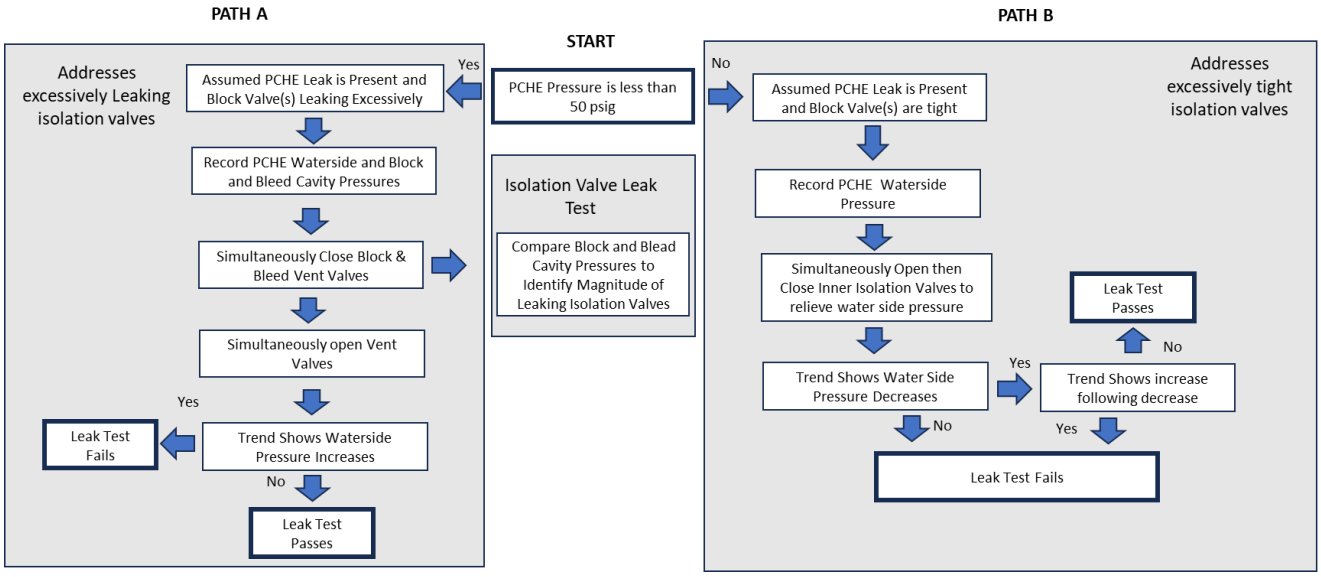
- **Path B**

  - IP PCHE water side pressure is recorded.
  - IP Inlet and Outlet isolation valves are simultaneously opened.
  - Valves are kept open for 3 seconds then command closed.
  - If IP water side pressure does not show a decrease of 20 psi during this period, the IP leak test FAILS.
  - If water side pressure shows a decrease, then and increase by at least 10 psi after the open to close process (10 seconds after valve detected closed), the IP leak test FAILS.
  - If the IP water side pressure does not show an increase of at least 10 psi after the valve open to close process (10 seconds after valve detected closed) the IP leak test PASSES.

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HP Section

- HP PCHE water side pressure as measured is measured. If HP pressure is less than 50 psig, **Path A** is executed. If pressure is greater than 50 psig, **Path B** is executed.



- **Path A**

  - HP Inlet and Outlet Double Block and Bleed cavity pressures are recorded.
  - HP Inlet and Outlet Double Block and Bleed Vent Valves are simultaneously closed.
  - Valves kept closed for 5 seconds then command open.
  - HP Inlet and Outlet Double Block and Bleed cavity pressures from the point of each vent valves being sensed closed to the point at which they begin to open are measured. If a pressure rise of 5 psi or greater is detected in either the IP Inlet Double Block and Bleed cavity or HP Outlet Double Block and Bleed cavity during this time the HP leak test FAILS.
    - A leak identified by Path A indicates one or both of the HP the water side isolation valves is leaking excessively. An ANUNCIATION is activated identifying which or if both valves are leaking excessively. Maintenance should be performed on these valves.
  - If a pressure rise of less than 5 psi is seen in either the HP Inlet Double Block and Bleed or HP Outlet Double Block and Bleed cavities is detected, the HP leak test PASSES.

- **Path B**

  - HP PCHE water side pressure is recorded.
  - HP Inlet and Outlet isolation valves are simultaneously opened.
  - Valves are kept open for 3 seconds then command closed.

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- If HP water side pressure does not show a decrease of 20 psi during this period, the HP leak test FAILS.
- If water side pressure shows a decrease, then and increase by at least 10 psi after the open to close process (10 seconds after valve detected closed), the HP leak test FAILS.
- If the HP water side pressure does not show an increase of at least 10 psi after the valve open to close process (10 seconds after valve detected closed) the HP leak test PASSES.

E. Any failed leak test shall be reported to GEV Product Service Engineering. Instructions can be provided on the method of restarting the GT following a failed leak test. If an excessive leak is present, there may problems in during the restarting process due to elevated amounts of water in the gas fuel.Low Water to Gas Pressure Protection

At all times during performance heater operation, it is critical that the water pressure be higher than gas pressure. Reducing the water pressure below the gas pressure in the presence of a PCHE leak can result in natural gas being sent to the condensate return system. Due to the nature of the HRSG system, gas can then migrate to a high point in that system which is typically not hazardous area rated. Additionally, there may be no method to detect gas in the system and it may pose a safety hazard if hot work is performed. In order to mitigate this hazard, the water pressure must be verified to be higher than gas fuel pressure at all times when the performance heater is active.

The feedwater inlet and outlet are in double block and bleed state when the feedwater pumps are not operating. This acts to prevent gas fuel from entering the feedwater system or condenser in the event of a heat exchanger leak.

During the performance heater Start-up Mode, once the feedwater inlet control valve (VA41-6) is at the minimum stroke position and the feedwater inlet bleed valve (VA41-8) is closed, the turbine control panel will monitor the inlet water pressure. No further action is taken until the water inlet double block and bleed cavity pressure transmitters (96PH-W1X) are confirmed to be higher than the gas inlet pressure (96PH-G1X).

During Normal Operation Mode, the water to gas pressure protection uses the feedwater outlet pressure transmitters (96PH-W4X) to compare outlet water pressure to the gas inlet pressure (96PH-G1) for all further operating modes. These locations provide conservative pressures are used on both streams – gas pressure monitoring on the inlet side (as to not account for any pressure drop across the heat exchanger) and water pressure is on the outlet side (accounts for pressure drop across the heat exchanger). If this feedwater pressure is determined to not be sufficiently higher than gas pressure at any point during operation, the performance heater will isolate feedwater through the inlet and outlet double block and bleed valves (see Shutdown State), the gas fuel inlet isolation valve (VA41-2) will close, and the gas fuel bypass valve (VA41-21) will open.

Redundant water outlet pressure transmitters are included on the gas fuel performance heater skid. If the pressure transmitters do not provide a reading, then the feedwater flow to the gas fuel performance heater skid will be isolated.

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NEVER DISABLE THE LOW WATER TO GAS FUEL PRESSURE PROTECTION SINCE IT IS USED TO DETECT GAS FUEL ENTERING THE FEEDWATER SYSTEM.

F. Bleed Valve Not Closed Protection

The gas fuel performance heater skid will be isolated from the IP and HP feedwater system (Shutdown State) if the bleed valves (VA41-7 & VA41-8) are out of position (not fully closed) to prevent water/steam from overflowing the drain tank.

If the position feedback signal from one or both of the bleed valves is lost or bad quality, then the feedwater system will be isolated (Shutdown State).

The HP Section will be isolated from the HP feedwater system (HP Section Isolation) if the HP bleed valves (VA41-71 & VA41-81) are out of position to prevent water/steam from overflowing the drain tank.

If the position feedback signal from one or both of the HP bleed valves is lost or bad quality, then the HP feedwater system will be isolated (HP Section Isolation).

G. Feedwater Pressure Oscillations

During commissioning of the gas turbine, prior to completing all control loop tuning, the gas fuel performance heater may be subject to oscillations in the feedwater supply pressure. When the feedwater pressure oscillates while the performance heater is active, the temperature control valve will begin modulating in an attempt to maintain the target gas fuel temperature. The changes in feedwater supply pressure and temperature control valve position will result in instability of the gas outlet temperature and subject the heat exchanger to an unintended operating condition. Due to this, it is recommended to conduct all feedwater system control loop tuning prior to commissioning the gas fuel performance heater.

H. IP Feedwater Return Temperature Protection

Due to concerns regarding flashing in the feedwater return system downstream of the performance heater skid, the feedwater flow to the performance heater will be isolated (Shutdown State), the gas fuel inlet isolation valve (VA41-2) will close, and the gas fuel bypass valve (VA41-21) will open if the feedwater outlet temperature measured by FT-PH-4X exceeds a maximum temperature limit. This protection is intended to prevent steam from being returned to the downstream system and requiring the downstream relief valve to be sized for steam flow which will increase the required capacity of the valve.

I. IP Temperature Control Valve Closed Protection

If the IP temperature control valve (VA41-9) position feedback signal indicates that VA41-9 is closed when the gas fuel performance heater is in operation, the IP and HP feedwater flow to the performance heater skid will be shut off (Shutdown State), the gas fuel inlet isolation valve (VA41-2) will close, and the gas fuel bypass valve (VA41-21) will open to prevent over-pressurizing the IP Section of the system with HP feedwater pressures.

J. High IP Feedwater Pressure Protection

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If the pressure at the IP Section feedwater inlet to the performance heater (as measured by 96PH-W61, 96PH-W1, and 96PH-W8) is detected to be close to the IP system design pressure, the HP Section will isolate (HP Section Isolation) to prevent over-pressurizing the IP Section of the skid prior to reaching the setpoint of the IP Section relief valve (VR15-4). The gas fuel will continue to be heated by the IP feedwater through the IP Section of the heat exchanger, however the gas outlet temperature will be reduced due to the loss of HP feedwater flow.

K. High HP Feedwater Outlet Temperature Protection

If the feedwater temperature at the HP outlet (measured by FT-PH-41X) is detected to be close to the IP system design temperature, the HP Section will isolation (HP Section Isolation) to prevent exceeding the design temperature of the IP Section of the heat exchanger skid. The gas fuel will continue to be heated by the IP feedwater through the IP Section of the heat exchanger, however the gas outlet temperature will be reduced due to the loss of HP feedwater flow.

L. Low IP Feedwater Flow Protection

If the IP feedwater flow rate at the IP inlet to the performance heater skid (measured by the IP inlet flow meter differential pressure measurements 96PH-W7X) is below the minimum IP flow setpoint, or if the signal from both differential pressure transmitters is lost, the HP Section of the heat exchanger will isolate (HP Section Isolation) to prevent loss of IP flow and potentially backflowing into the IP feedwater supply system. The gas outlet temperature will be reduced due to the loss of HP feedwater flow.

V. DESCRIPTION OF HMI SCREEN BUTTONS

A. GFPH Control: ON/AUTO

Push buttons are provided to enable the performance heater once the gas turbine is operational. When the gas turbine is fired, the control is automatically turned on and cannot be turned off. In the event of a performance heater trip while the gas turbine is operational, refer to the separate section in this document regarding the Shutdown State.

B. GFPH Control: OFF

If the gas turbine is started with the heater “OFF” push button selected, then the gas turbine will operate on cold fuel, and the performance heater cannot be turned on while the gas turbine is operational. The unit must be shut down and on turning gear to enable the performance heater.

During gas turbine operation, the performance heater cannot be turned off via this pushbutton. If it is selected during gas turbine operation, the heater will continue operating until the gas turbine is shutdown. If the push button is still “OFF” during the next gas turbine start, then it will not be started.

VI. SYSTEM REQUIREMENTS

A. Instrument Air Supply

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The pneumatic valves on the gas fuel performance heater skid are configured per the requirements in GEK110727. Stroke all the automated valves on the skid to confirm they are actuating properly prior to operating the performance heater skid. The feedwater isolation and control valves are configured to close on loss of instrument air pressure and the feedwater bleed valves will open on a loss of instrument air pressure. Instrument air pressure to the valves should be checked to avoid the valves reverting to their de-energized position and result in feedwater supply being shut off to the PCHE.

B. Feedwater Cleanliness

The gas fuel performance heater skid has filters, one for the IP Section and one for the HP Section, installed at the inlet to the PCHE to protect it from particulate in the feedwater. However, it is still important that the feedwater system be thoroughly cleaned per the requirements in GEK110483 prior to introducing feedwater flow to the performance heater skid. Inefficient cleaning practices for the feedwater system can increase fouling of the heat exchanger and/or impact performance of the water side valves.

Feedwater cleanliness is important since the gas turbine must be shutdown to replace the feedwater filter cartridge on the performance heater skid if it becomes clogged. Online replacement of the permanent filter cartridge is not advised. Ensuring that the feedwater is clean will help extend the filter life and reduce gas turbine downtime for filter maintenance.

Pressure instrumentation on the skid is used to measure the differential pressure across the heat exchanger to monitor for clogging or fouling. Refer to the PCHE maintenance section for cleaning recommendations if high PCHE differential pressure is detected.

Confirm proper installation of the conical start-up strainer in the feedwater line prior to introducing feedwater to the performance heater skid. The point of the conical strainer should be directed against water flow and the mesh should be on the outside of the cone. Confirm filter cartridges are installed in the feedwater filter housings prior to commissioning. Spare feedwater filter cartridges are recommended to be kept on-site in case the filter need to be replaced.

Debris in the feedwater system can result in feedwater isolation and bleed valve issues. Debris stuck in the seat of the bleed valves can cause them to not fully close which will result in feedwater being isolated (Shutdown State) or excess leaking of feedwater when the unit is shutdown. If the bleed valves are not fully closing, then it is recommended that the gas turbine be operated with the performance heater “OFF” until the bleed valves can be repaired or replaced to avoid repeatedly entering a Shutdown State Condition and exposing the PCHE to harsh thermal transient events.

C. Gas Fuel Cleanliness

The gas fuel system shall be cleaned per the requirements of GEK110483 prior to introducing gas fuel flow to the performance heater skid. Refer to GEK116771 for recommendations on how to clean the gas fuel system piping. The gas fuel supply shall be per GEI41040.

Pressure instrumentation is used to measure the differential pressure across the gas side of the PCHE to monitor for clogging or fouling. Refer to the PCHE maintenance section for cleaning recommendations if high PCHE differential pressure is detected.

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D. Vent and Drain Line Routing

Some water side vents and drains are required to be routed to a hazardous area rated atmospheric drains tank. Valve discharges with this requirement are:

1. VA41-8: IP feedwater side inlet double block and bleed cavity vent
1. VA41-7: IP feedwater side outlet double block and bleed cavity vent
2. VR15-2: IP Feedwater side relief valve
3. VA41-81: HP feedwater side inlet double block and bleed cavity vent
4. VA41-71: HP feedwater side outlet double block and bleed cavity vent
5. VR15-21: HP feedwater side relief valve

The water side pressure relief valve and water side double block and bleed vents have the potential for discharging natural gas in the event of a heat exchanger leak. When a heat exchanger leak is present, any leaking gas during a shutdown (when the feedwater is isolated from the heat exchanger) will exit through the vents and/or drains rather than migrating throughout the water cycle. Over time water can leak across the double block and bleed isolation valves, resulting in the isolated water pressure dropping below the gas pressure. In the event of a leak in the heat exchangers, this can result in gas migrating into the water side of the heat exchanger skid and going through the drain connection. As a precaution, it is required that the feedwater drains be routed to a tank that is rated for such a mixture.

VR15-4 (HP feedwater outlet to IP feedwater inlet relief valve) does not have the requirement to be routed to a hazardous area rated atmospheric drain tank as it is outside of the double block and bleed cavity configured to protect from gas migration to this valve.

The water side double block and bleed cavities are equipped with a restriction orifice prior to entering the drain tank. The orifice is used with an upstream pressure transmitter (96PH-W10A for HP drain monitoring and 96PH-W10B for IP drain monitoring) and the drain valve positioners (VA41-71 and VA4-81 for HP side drains; VA41-7 and VA41-8 for IP side drains) to confirm the position of the valve. This method results in a “Shutdown State”, the gas fuel inlet isolation valve (VA41-2) closing, and the gas fuel bypass valve (VA41-21) opening when drain line pressure transmitter reaches the required threshold and the corresponding positioner confirms the valve is not closed.

Drain line orifice and pressure transmitters are not shown in Figure 1 for clarity. For further information, refer to the project specific schematic (MLI 4002).

E. FOUNDATION™ Fieldbus System

The automated feedwater valves on HA-frame performance heater skids have digital valve controllers that utilize FOUNDATION™ Fieldbus protocol. The feedwater isolation and control valves are configured to close on loss of communication with the turbine control panel. The FOUNDATION™ Fieldbus communication between the valves and the turbine control panel should be properly commissioned without interference to avoid having the valves revert to their de-energized position and result in feedwater supply being shut off to the PCHE. This can result in damage to the PCHE.

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F. Return System Over Pressure Protection

The gas fuel performance heater does not provide any method for over pressure protection for the feedwater supply or return systems connected to the performance heater skid. The relief valves on the heat exchanger protect the heat exchanger during shutdown periods. Similarly, the temperature control valve is used to control the gas fuel outlet temperature of the performance heater. Neither of these are designed to limit the return pressure from the performance heater skid. Due to this, a dedicated over pressure protection method must be supplied (outside the scope of the skid) to protect the return system from exceeding the design pressure.

G. Freeze Protection

The PCHE, water piping, and water drains must be heat traced and insulated on all projects where the site ambient temperature can drop below freezing. The gas fuel supply temperature to the performance heater skid shall always be at least 40 degrees Fahrenheit to prevent the water in the PCHE from freezing when there is gas flow through the PCHE but no feedwater flow.

The heat exchanger, all gas fuel piping downstream of the heat exchanger, and the feedwater piping must be insulated to reduce heat loss and protect site personnel.

VII. MAINTENANCE REQUIREMENTS

A. IP Feedwater Inlet Temporary Strainer

The gas fuel performance heater skid has a temporary strainer installed at the feedwater inlet to the performance heater skid for use during initial commissioning to help protect the feedwater valves and the PCHE from particulate in the feedwater.

The temporary water strainer is a conical type strainer with mesh on the upstream side for catching particulates before they reach the PCHE core and/or permanent filter. The strainer is installed between flanges in the piping, with a removable pipe spool to allow disassembly for cleaning or replacement. A differential pressure gauge is provided across the water side strainer to identify if the strainer has become contaminated and needs cleaning.

The water side temporary strainer shall remain installed for at least 100 hours of GT operation (with feedwater flowing) and be removed prior to reaching 200 hours of operation. The differential pressure indicator shall be checked frequently when there is feedwater flow to the performance heater to determine if it has become clogged. If the differential pressure across the strainer exceeds 50 psid, the gas turbine shall be shut down, and the strainer shall be removed, cleaned, and re-installed.

B. Gas Inlet Temporary Strainer

A temporary strainer at the gas fuel inlet to the PCHE is provided on the gas fuel performance heater skid. The temporary gas strainer is a conical type strainer with mesh on the upstream side for catching particulates before they reach the PCHE. The strainer is installed between flanges in the piping, with a removable pipe spool to allow for easier disassembly for cleaning or extraction. The gas side strainer shall remain installed for at least 10 hours of GT operation and be removed prior to reaching 200 hours of operation. Differential pressure monitoring across the strainer is not required.

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C. Feedwater Inlet Permanent Filter

The gas fuel performance heater skid has a permanent filter installed at the HP and IP feedwater inlet to the PCHE to protect it from particulate in the feedwater. The intent of the permanent feedwater filter is to provide the final filtration required prior to feedwater entering the PCHE, also acting as a material transition for the feedwater piping from carbon steel to stainless steel. The mesh size of the permanent filter is sized based on the filtration requirements as determined by the PCHE manufacturer.

The permanent filter has differential pressure transmitters (96PH-W2 and 96PH-W21) installed to monitor the pressure drop across the filter. An alarm will annunciate when the differential pressure is high to alert the operator that the filter should be changed the next time the gas turbine is shut down. If the differential pressure across the filter reaches the high-high level, an alarm will annunciate to alert the operator that the gas turbine needs to be shut down as soon as possible to change the filter.

Continuing to run after the high-high differential pressure is reached risks collapsing the filter cartridge and damaging the PCHE and downstream valves with debris. There is sufficient margin between the high level and high-high level setpoints to give the operator time to plan a gas turbine shutdown after the high-level alarm differential pressure setpoint is reached if there are moderate levels of particulate in the feedwater supply.

CAUTION

If the differential pressure across the filter rises quickly or if frequent filter cleanings are required, it is indicative that the feedwater supply has not been properly cleaned and needs to be addressed prior to resuming performance heater operation.

For projects with a simplex filter arrangement, the gas turbine must be shutdown to clean the filter.

Some performance heater skids include duplex filters vessels. The duplex filter arrangement is intended to allow an on-line filter changeover to extend the operational time of the heat exchanger prior to shutting down the unit. Refer to the manual provided by the filter manufacturer for the procedure to follow for online transfer to the standby (clean) filter vessel. It is recommended that the gas turbine be shutdown prior to changing the filter cartridges even with duplex filter vessels.

WARNING

IF PERFORMING ONLINE MAINTENANCE OF THE STANDBY FILTER VESSEL, SITE SAFETY PROCEDURES SHALL BE FOLLOWED TO AVOID POTENTIAL EXPOSURE TO HIGH PRESSURE, HIGH TEMPERATURE FEEDWATER.

The following procedure is intended to provide general guidance to replace the feedwater filter cartridges. This section is not intended to override GEKs, site safety procedures, or the

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filter manufacturer replacement procedure. This procedure is a recommended sequence to change out the filter cartridges and verify that all equipment is returned to the normal position to enable operation during the next startup.

As an additional precaution, it is recommended that the gas side of the heat exchanger be isolated and vented so that in the event of an unidentified heat exchanger leak, there is no risk of gas migrating from the gas side of the heat exchanger to the water side where personnel are present.

To isolate the gas side of the performance heater, the following procedure is recommended but may be tailored to align with site specific safety guidelines. This is intended to a representative sequence of events to be followed:

1. The gas inlet double block and bleed valve (VM-PH-1) shall be closed so that the gas side is isolated.

2. The gas outlet double block and bleed valve (VM-PH-2) shall be closed so that the gas side is isolated.

3. The bleed valve and/or plug shall be opened/removed from the bleed port on VM-PH-1 and VM-PH-2 so that any gas leaking by the isolation valve is discharged to atmosphere and not to the heat exchanger.

4. The gas side manual vent valve shall be opened to depressurize the heat exchanger cavity.

Once the gas side has been isolated, the water system shall be isolated, and the filter cartridge can be replaced per the below procedure.

1. The filter housing must be isolated from the water system. It is recommended that the gas turbine is shutdown and the feedwater pumps are shutdown when the filters are being replaced.

    a. Isolation valves upstream and downstream of the filter vessel are fully closed.

    b. Bleed valves in the feedwater double block and bleed cavities are fully open.

    c. Verify the filters are depressurized by checking the pressure transmitter that reads the feedwater pressure in the piping section connected to the filter vessel.

    d. Appropriate lock-out-tag-out procedures shall be followed to make sure that the isolation valve and bleed valves are not actuated during the filter change out. The instrument air supply isolation valves upstream of the actuators should be closed to return their valves to their de-energized state.

2. Once the filter housing is isolated from the system and depressurized, the housing can be drained. Open the filter vessel drain valve.

3. After the filter vessel has been drained and it has been confirmed that the pressure inside the filter vessel is atmospheric pressure, loosen the filter cover bolts and remove the cover.

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4. Remove the existing filter cover gasket and inspect the filter cover for signs of damage.

5. There may be nuts or a hold down cap holding the filter cartridge in place that need to be removed to remove the cartridge from the filter vessel. Keep the nuts or filter hold down cap that held the cartridge in place. Discard the existing filter cartridge.

6. Install the new filter cartridge in the filter vessel. The cartridge should be properly aligned in the filter vessel so no water can bypass the filter cartridge. Reinstall the nuts or hold down cap holding the cartridge in place.

7. Clean the filter cover and sealing surface. Inspect the sealing surface and gasket groove for nicks that could result in a leakage path. Examine the new filter cover gasket to confirm there are no indications of damage or debris. Install the new filter cover gasket in the filter cover. Place the filter cover back on the filter vessel.

8. Use a calibrated torque wrench to tighten the cover bolts per the manufacturer's installation and operation manual. Bolts shall be lubricated with a low friction moly anti-seize. Use care to avoid galling when tightening stainless steel bolting.

9. Close the filter vessel drain valve and reopen all the instrument air isolation valves going to the pneumatic automated valves.

10. Complete pre-filling and venting of the filter vessel.

11. Check the filter cover for leaks.

After all maintenance activities have been completed and the water side of the skid is ready to be returned to service, the following steps shall be followed to prepare the gas side for operation.

1. The PCHE gas side manual vent valve shall be closed.

2. If any temporary vent lines were installed for the gas side double block and bleed valves (VM-PH-1 and VM-PH-2), they shall be removed, and the plug shall be installed on the bleed port of the valves.

3. After installing the plug on the bleed ports of the valves, the inlet isolation valve (VM-PH-1) shall be slowly opened allowing a gradual pressurization of the cavity downstream of the valve.

4. After the inlet isolation valve (VM-PH-1) is fully opened, the outlet isolation valve (VM-PH-2) shall be slowly opened allowing a gradual pressurization of the cavity downstream of the valve.

During the next startup, when water is introduced, an initial service leak test shall be performed to determine any leaks present in the system as a result of the maintenance activities. Since no hot-work has been considered as part of this modification, full hydrostatic testing is not required. However, due to the risk of potential injury, the skid shall be monitored from an appropriate distance in the event external leaks are present.

WARNING

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DRAIN AND VENT THE FILTER VESSEL TO DEPRESSURIZE IT PRIOR TO SERVICING THE FILTERS. REFER TO THE MANUFACTUER'S MANUAL FOR DETAILED INFORMATION. USE CAUTION IF CHANGING FILTER CARTRIDGES DURING LIQUID FUEL OPERATION.

VIII. PCHE COMISSIONING AND MAINTENANCE

A. PCHE Cleaning

Both the gas and water sides of the PCHE core have differential pressure measurements, which are used by the gas turbine controller to monitor and report on fouling and/or blockage of the PCHE core passages to allow the operator plan for cleaning of the PCHE core at their next outage. An alternate indication of fouling that may require PCHE maintenance is a significant increase in feedwater water flow through the skid to maintain the target gas fuel temperature compared to the water flow required when the PCHE was commissioned.

The PCHE skid core has been provided with numerous maintenance passages, which provide access to the water and gas header sections. These maintenance passages can be identified as blind flanged connections on the sides of the water and gas headers (inlet and outlet), with flange faces perpendicular to the primary flow paths. See the PCHE arrangement drawing for more information for the location of the maintenance ports.

When cleaning the PCHE core, the following steps should be followed:

1. Create a maintenance plan and procure the necessary equipment before beginning the cleaning process. This includes replacement flange gaskets for the maintenance ports.

2. Remove the blind flanges and insert a borescope or similar device into the headers, to identify the location, extent, and make-up of the blockages. Generally, the blockages will develop at the inlets to the PCHE core passages, at the extreme ends of the headers.

3. It is highly recommended to contact the PCHE core manufacturer for advice and information to assist in the cleaning of the system. To get the most benefit from their assistance, it is best to be prepared with knowledge of the material that is causing the blockage and the extent of the blockage.

4. There are several methods for cleaning the core, which are explained below. Approved cleaning methods shall be verified through the PCHE maintenance manual.

    a. Back Puffing – Puffing a high-pressure gas to expel or loosen debris.

    b. Back Flushing – Flowing water in the reverse direction of normal flow.

    c. Ultra-High-Pressure Water Jetting – This method requires trained and qualified operators, who will backward and forward flush the system with extremely high pressure, high velocity water. This is the most effective method of cleaning.

    d. Chemical Cleaning – This method required trained and qualified operators. Please contact the PCHE core manufacturer for recommended cleaning chemicals and/or companies to perform this form of maintenance.

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B. PCHE Maintenance Flanges

During initial commissioning and after maintenance (if the maintenance flanges are removed), the maintenance flange bolts will need to be torqued by the site. Refer to ASME PCC-1 guidance on torquing bolts for ANSI flanges and refer to 249A8350 for guidance on installing and torquing quick disconnect flanges if information is not provided in the PCHE manual. If torque values are provided in the PCHE manual, then they take precedence over the values in ASME PCC-1 or 249A8350 and the values in the PCHE manual shall be used instead.

C. PCHE Sliding Supports

During initial commissioning and during outages, check that proper torque has been applied to the sliding support hold-down bolts of the PCHE. Refer to the PCHE manual for the proper torque values. This is critical so the heat exchanger can expand thermally without binding, which can be caused by an over-tightened bolt.

D. PCHE Removal

Refer to 319T6816 for the procedure to remove the PCHE from the gas fuel performance heater skid.

IX. REFERENCES

1) GEI41040 – Specification for Fuel Gases for Combustion in Heavy-Duty Gas Turbines
2) 319T6816 – COMBINED IP AND HP HEAT EXCHANGER REMOVAL PROCESS
3) GEK110483 - Cleanliness Requirements for Power Plant Installation-Commissioning and Maintenance
4) GEK116771 – Gas Fuel Supply System Cleaning for Gas Turbines
5) GEK110727 - Plant Instrument Air System Requirements
6) GER-4189 - DESIGN CONSIDERATIONS FOR HEATED GAS FUEL
7) GEK111309 – GE VERNOVA Product Safety Recommended Best Practices
8) MLI 4002 Gas Fuel Performance Heater Schematic (Project Specific)
9) MLI 0422 Gas Fuel System Schematic (Project Specific)
10) MLI 4063 Customer Connection List (Project Specific)
11) MLI 0414 Instrument List (Project Specific)
12) 249A8350 Clamped Flange Connector Instructions

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